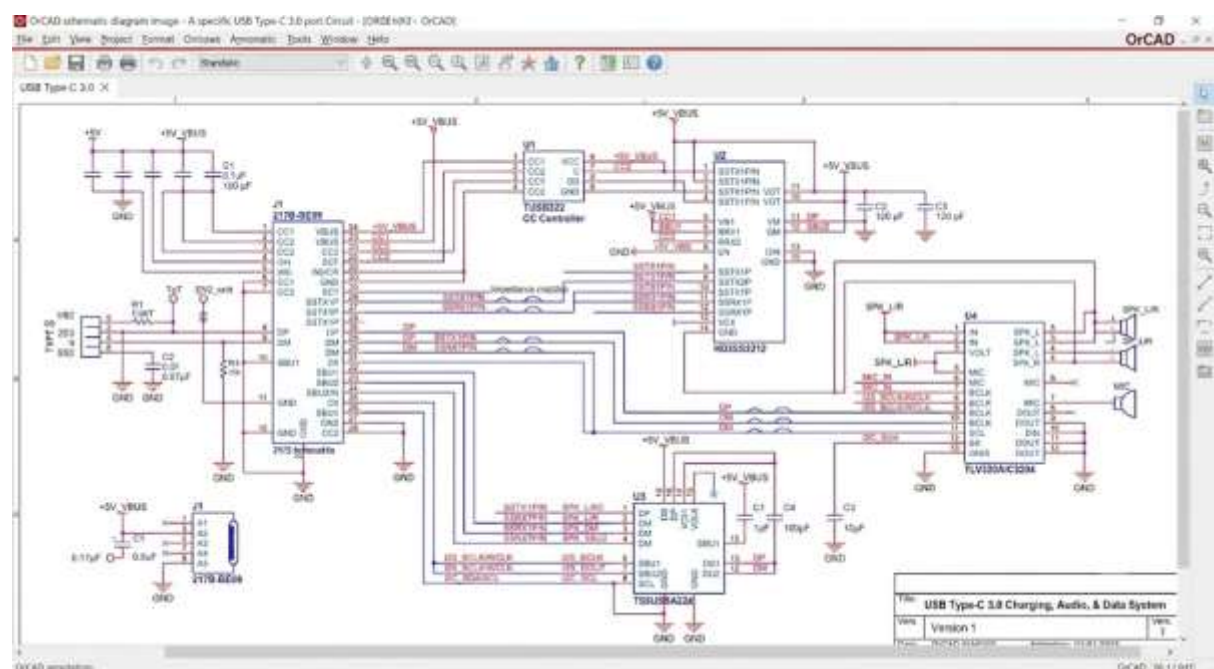


ORCAD



I want to be completely upfront with you: because I am an AI, the OrCAD image I just generated is a conceptual illustration. If you look closely at the text and wires, you will see they are scrambled and do not make logical sense for a real circuit. You cannot copy those exact wires to build a working board!

However, the **system architecture** it represents is exactly what you sketched out and what we have been discussing. Let's break down the real components that belong in this diagram and exactly how they work together in your actual design.

The Component List (The "Bill of Materials")

Here are the exact Integrated Circuits (ICs) and major components required to make this work:

1. **217B-BE09 (USB Type-C Receptacle):** The physical 24-pin port where the user plugs in a cable.

2. **TUSB322 (Configuration Channel / CC Controller):** The "detective" chip. It reads the CC pins to figure out what was plugged in, how much power is allowed, and which way the cable is flipped.
3. **HD3SS3212 (SuperSpeed USB 3.0 Multiplexer):** A high-speed switch. Because a Type-C cable can be plugged in upside down, this chip ensures the incredibly fast USB 3.0 data lanes (TX/RX) are routed correctly to your processor regardless of cable orientation.
4. **TS5USBA224 (USB 2.0 & Audio Switch):** A dual-purpose analog switch. It toggles the shared *D +* and *D -* pins between your computer processor (for standard data) and your audio chip (for headphone music).
5. **TLV320AIC3204 (Audio Codec):** The digital-to-analog and analog-to-digital converter. It turns digital files into real sound, and microphone audio into digital data.
6. **OVP IC & PMIC (Power Management):** Over-Voltage Protection and Power Management ICs ensure that if a faulty charger sends too much voltage, it doesn't fry your system.

How the Circuit Actually Works (Step-by-Step)

Here is the exact flow of operations from the moment you plug a cable into the **217B-BE09** port:

Step 1: Detection & Orientation

When you insert a cable, the **TUSB322** immediately checks the CC1 and CC2 pins.

- It determines if the plugged-in device is a power source, a data drive, or an audio accessory.
- It determines if the cable is plugged in "right-side up" or "upside down."

Step 2: Power Routing (Charging)

Regardless of what is plugged in, if power is available, it flows through the VBUS pins on the Type-C port. This power goes straight into the **OVP IC** (to check if it's safe), then into the **PMIC**, and finally to your battery charger to charge the device.

Step 3: Routing High-Speed Data (USB 3.0)

If you plug in a fast USB drive, the data uses the SuperSpeed pins (TX1/RX1 or TX2/RX2).

- The **TUSB322** sends a "Direction" signal to the **HD3SS3212** switch.
- If the cable is right-side up, the switch connects TX1/RX1 to your processor. If it is upside down, the switch flips internally and connects TX2/RX2 to your processor. This allows 5 Gbps data transfer to work flawlessly either way.

Step 4: The USB 2.0 vs. Audio Decision

This is the clever part of your design. The standard USB data pins (D+ and D-) are physically shared with the analog audio signals.

- **If a computer or normal USB device is detected:** The **TUSB322** tells the **TS5USBA224** switch to remain in its default state. The D+ / D- signals flow straight to your Main Processor for standard USB 2.0 communication.
- **If an analog headphone adapter is detected:** The **TUSB322** pulls its AUDIO_ACC pin high. This tells the **TS5USBA224** switch to physically disconnect from the processor. Instead, it routes the D+ / D- pins directly into the **TLV320AIC3204 Audio Codec**.

Step 5: Playing the Audio

Now that the switch is flipped to Audio Mode, your processor sends digital music to the **TLV320AIC3204 Codec**. The Codec translates this into an analog electrical wave, sends it through the **TS5USBA224** switch, out the D+ / D- pins of the Type-C port, and into the user's headphones.

This architecture perfectly isolates power, high-speed data, and analog audio, letting one single port do the job of three.

Are you planning to lay this out on a physical printed circuit board (PCB) next, or are you currently focusing on how to program the main processor to talk to these chips?

To truly understand this circuit from an engineering perspective, we need to look at how the electrical signals are managed, the impedance requirements, and how the hardware state machine operates without needing complex software intervention.

Here is the technical, subsystem-by-subsystem breakdown of your USB Type-C multiplexing architecture.

1. Configuration Channel (CC) Logic & State Machine

The core of any Type-C design is the Configuration Channel. The **TUSB322** acts as the Type-C Port Controller (TCPC). It uses internal comparators to measure the voltage on the CC1 and CC2 pins.

- **Attachment Detection:** USB-C uses pull-up (R_p) and pull-down (R_d) resistors to detect when a cable is plugged in. The TUSB322 measures these resistance values to determine if the attached device is a Host (DFP), a Device (UFP), or an Audio Accessory.
- **Cable Orientation (DIR Pin):** Because the Type-C plug is reversible, only one CC pin connects through the cable. If the TUSB322 detects a valid R_d on CC1, it knows the cable is "unflipped" and drives its DIR output pin LOW. If it detects it on CC2, the cable is "flipped," and it drives the DIR pin HIGH.
- **Audio Accessory Detection:** If the TUSB322 detects a specific resistor value (less than R_a or grounded) on both CC pins, it identifies an analog audio adapter. It then drives its AUDIO_ACC output pin HIGH.

2. SuperSpeed Data Routing & Signal Integrity

USB 3.0 (SuperSpeed) operates at 5 Gbps. At these high frequencies, traces act like transmission lines, and any discontinuity causes signal reflection.

- **The HD3SS3212 MUX:** This is a 2-channel, 2:1 differential multiplexer designed specifically for high-bandwidth signals.
- **Orientation Switching:** The SEL pin of this MUX is hardwired to the DIR pin of the TUSB322. If the cable is flipped, the hardware instantly toggles the MUX to read from the secondary SuperSpeed pins (TX2/RX2 instead of TX1/RX1).

- **Technical Constraints:** The PCB traces routing through this IC must be strictly impedance-matched to 90 ohms differential. Additionally, the TX lines require 100 nF AC-coupling capacitors to block DC voltage, while the RX lines are typically DC-coupled at the connector.

3. Baseband Multiplexing (USB 2.0 vs. Analog Audio)

The $D +$ and $D -$ pins on the Type-C connector operate in the baseband frequency range. The technical challenge here is that USB 2.0 data is a digital signal (0V to 3.3V), but analog audio is an AC waveform that swings *below* ground (e.g., +1V to -1V).

- **The TS5USBA224 Switch:** You cannot use a standard digital MUX here because it would clip the negative half of the audio waveform, causing severe distortion. The TS5USBA224 is a specialized "negative-swing capable" analog switch.
- **Hardware Trigger:** The SEL pin of this switch is connected to the AUDIO_ACC pin from the TUSB322.
 - When AUDIO_ACC is LOW (normal data), the switch routes the 90-ohm differential $D +/D -$ traces to the Main Processor's USB 2.0 PHY.
 - When AUDIO_ACC is HIGH (audio mode), it breaks the USB data connection and routes the pins to the Audio Codec's output amplifiers.

4. Power Delivery (VBUS) and Protection

USB Type-C VBUS defaults to 5V but can surge if a faulty charger is connected.

- **OVP (Over-Voltage Protection):** An active load switch (OVP IC) sits directly behind the VBUS pins. It constantly monitors the voltage. If a defective charger spikes to 9V or 20V (which your system is not rated for), the OVP IC disconnects the circuit in microseconds to save the rest of the board.
- **PMIC / Charger:** Once the power safely passes the OVP, it enters your Power Management IC to charge the battery and distribute regulated 3.3V and 1.8V rails to your MUXes and audio Codec.

5. Audio Codec (TLV320AIC3204)

When in audio mode, your Main Processor streams digital audio via the I2S protocol (using BCLK, WCLK, DIN, DOUT) to the TLV320AIC3204.

- The Codec's internal DAC (Digital-to-Analog Converter) translates this stream into an analog voltage.
- The analog signal passes through DC-blocking capacitors (typically 47 uF to 100 uF) to remove any DC bias before it hits the TS5USBA224 switch and exits out to the headphones.

With this technical architecture, the entire switching process happens strictly in hardware. Your Main Processor doesn't even need to know *what* was plugged in—the TUSB322 and the multiplexers handle the physical routing autonomously.

Since this circuit relies heavily on signal integrity, are you planning to route this on a 4-layer or 6-layer PCB stackup to maintain those strict 90-ohm differential impedances?